AGA KHAN UNIVERSITY EXAMINATION BOARD
SECONDARY SCHOOL CERTIFICATE CLASS IX EXAMINATION

Chemistry Paper I (Theory) — Annual 2023

Time: 1 hour 10 minutes | Total Marks: 40

INSTRUCTIONS: Answer all forty multiple choice questions. Choose ONE best answer for each question.

Q.1 The investigation of chemical changes in diseases, drug action and nutrition is carried out by a/ an

A. biochemist.

B. physical chemist.

C. industrial chemist.

D. inorganic chemist.

Q.2 In Ca3(PO4)2, the valencies of calcium and phosphate ions respectively are

A. 3 and 2

B. 3 and 4

C. 2 and 3

D. 2 and 4

Q.3 The mass of 2.5 moles of H2O is (Note: 16_O and 1_H)

A. 45.0 g.

B. 42.5 g.

C. 20.5 g.

D. 18.0 g.

Q.4 The number of molecules in 2.1 moles of carbon dioxide is

A. 2.87 × 10^22

B. 2.87 × 10^23

C. 1.26 × 10^24

D. 1.26 × 10^25

Q.5 A phenomenon that ALWAYS occurs in a chemical reaction is that it

A. liberates a gas.

B. produces precipitates.

C. forms a new substance.

D. shows a change in colour.

Q.6 Nichrome is prepared by combining

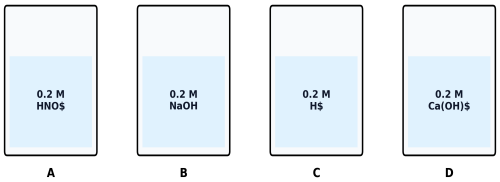
A. nickel and copper.

B. iron and chromium.

C. iron, nickel and copper.

D. iron, nickel and chromium.

Q.7 Which of the following is a diluted solution? (Note: Molar mass of HNO3=63, NaOH=40, H2SO4=98, Ca(OH)2=74 g/mol)



A. 0.2 M HNO3

B. 0.2 M NaOH

C. 0.2 M H2SO4

D. 0.2 M Ca(OH)2

Q.8 The concentration of a solution containing 53 g of Na2CO3 in 1 dm3 is (Note: Na=23, C=12, O=16)

A. 0.5 M.

B. 0.96 M.

C. 1.03 M.

D. 2 M.

Q.9 The technique that CANNOT separate the components of a homogeneous mixture is

A. filtration.

B. distillation.

C. crystallisation.

D. chromatography.

Q.10 A true solution is a homogeneous mixture of two or more substances. Which of the following is an example of a true solution?

A. Fog

B. Blood

C. Sugar syrup

D. Whipped cream

Q.11 The loss or gain of a particle that does NOT affect the mass of an atom is a/ an

A. proton.

B. neutron.

C. electron.

D. alpha particle.

Q.12 An ion Y+ has 39 nucleons and 18 electrons. The number of protons and neutrons that the nucleus of this ion contains is

A. 20 protons, 19 neutrons

B. 19 protons, 20 neutrons

C. 21 protons, 18 neutrons

D. 18 protons, 21 neutrons

Q.13 How many nucleons are present in an isotope of carbon (13_6 C)?

A. 13

B. 12

C. 7

D. 6

Q.14 The limitation of Rutherford atomic model is that the

A. mass and charge of an atom is concentrated in the nucleus.

B. positively charged nucleus resides in the centre of the atom.

C. electrons revolve around the nucleus at a high speed in fixed orbits.

D. revolving electrons continuously loose energy and fall on the nucleus.

Q.15 The electronic configuration of aluminium ion in aluminium oxide, Al2O3, is (Note: 27_13 Al)

A. 1s2 2s2 2p6

B. 1s2 2s2 2p6 3s1

C. 1s2 2s2 2p6 3s2 3p1

D. 1s2 2s2 2p6 3s2 3p2

Q.16 The STRONGEST oxidising ability among halogens is possessed by

A. fluorine.

B. chlorine.

C. bromine.

D. iodine.

Q.17 The general electronic configuration of ns2 np3 indicates that the element belongs to

A. period 3.

B. period 5.

C. group VA.

D. group IIIA.

Q.18 Consider the given characteristics: Soft, Low melting point, Highly electropositive. Based on these characteristics, the group of elements is identified as

A. halogens.

B. alkali metals.

C. carbon family.

D. oxygen family.

Q.19 The number of shells in the atomic structure of an element represents

A. its ionisation potential.

B. the group it belongs to.

C. the period it belongs to.

D. its electrical conductivity.

Q.20 The term, 'electronegativity' is described as the ability of an atom to

A. act as an oxidising agent.

B. accept an electron from another atom.

C. attract shared pair of electrons to itself.

D. absorb maximum energy to lose an electron.

Q.21 Water vapours have a lower density than ice because in water vapours the particles have

A. orderly arrangement.

B. vibrational movement.

C. very low kinetic energy.

D. very weak attractive forces.

Q.22 In a closed system, a change in kinetic energy of particles causes the substance to change its

A. state.

B. mass.

C. volume.

D. boiling point.

Q.23 During extremely cold weather when temperature is -20°C and room is at +20°C with 60% humidity, water drops and ice on the window result from

A. sublimation.

B. evaporation.

C. precipitation.

D. condensation.

Q.24 The term, 'allotrope' describes the difference in elements based on their

A. structural forms.

B. number of H+ ions.

C. number of neutrons.

D. phases (solid, liquid or gas).

Q.25 In industries, graphite is used as a/ an

A. electrode.

B. ornament.

C. electrolyte.

D. cutting tool.

Q.26 The compound that possesses ionic bond between its atoms is

A. CS_2

B. CCl_4

C. HCN

D. MgF_2

Q.27 In a reaction between chlorine (Group VIIA) and beryllium (Group IIA), each atom of chlorine

A. gives an electron to an atom of beryllium.

B. shares an electron with an atom of beryllium.

C. accepts an electron from an atom of beryllium.

D. shares an electron pair with an atom of beryllium.

Q.28 Hydrogen can form an ionic compound with

A. carbon.

B. sodium.

C. fluorine.

D. nitrogen.

Q.29 HCl is a polar covalent compound showing all of the following properties EXCEPT that it

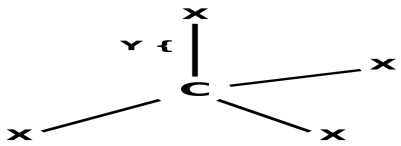
A. is soluble in water.

B. ionises completely in aqueous medium.

C. has partial positive and negative charges.

D. shows electronegativity difference as zero.

Q.30 In the tetrahedral structure where four species X connect to carbon by Y bonds, species X and Y bonds are:



A. atoms, weak

B. atoms, strong

C. ions, weak

D. ions, strong

Q.31 The oxidation state of chromium in CrO_3 is

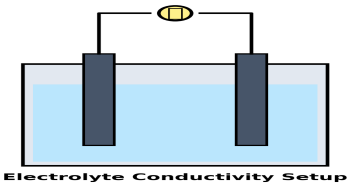
A. -6

B. -3

C. +3

D. +6

Q.32 High conductivity of an electrolyte in solution indicates that the substance tested must have been



A. petrol.

B. glucose.

C. lithium fluoride.

D. carbon tetrachloride.

Q.33 Metal electrodes are good conductors of electricity because they

A. contain partial charges.

B. are highly electronegative.

C. are connected to the battery.

D. contain delocalised electrons.

Q.34 A voltaic (galvanic) cell is an electrochemical cell which

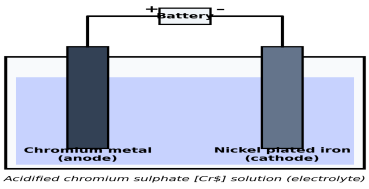
A. converts electrical energy into chemical energy.

B. generates electricity through spontaneous redox reaction.

C. comprises of two electrodes made up of the same inert metal.

D. consists of a positively charged anode and a negatively charged cathode.

Q.35 In the electrolytic cell used for electroplating iron with chromium, the cathode reaction is



A. Cr(s) -> Cr2+(aq) + 2e-

B. Cr2+(aq) + 2e- -> Cr(s)

C. Cr(s) -> Cr3+(aq) + 3e-

D. Cr3+(aq) + 3e- -> Cr(s)

Q.36 A metal oxide which can act as either an acid or a base to produce salt and water (amphoteric oxide) is

A. SiO2

B. CaO

C. Li2O

D. Al_2O_3

Q.37 Amongst the given group I elements, the first ionisation energy would be the HIGHEST for

A. lithium.

B. sodium.

C. rubidium.

D. potassium.

Q.38 In contrast to sodium, iron is a hard metal because it

A. requires greater energy to break.

B. has weaker metallic bonding.

C. belongs to group IIA.

D. is highly reactive.

Q.39 Consider: Solid, Brittle, Non-lustrous. The element described by these characteristics is

A. carbon.

B. copper.

C. oxygen.

D. aluminium.

Q.40 The elements of group IIA are known as

A. noble metals.

B. alkali metals.

C. transition metals.

D. alkaline earth metals.